

SINA YARI

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I am a Ph.D. student and graduate teaching/research assistant in Computer Science at George Mason University (expected May 2028), available to start an internship, research collaboration, or full-time opportunity immediately. Over the past several years, I have designed optimization- and security-aware scheduling techniques for safety-critical real-time embedded systems, including work to mitigate schedule-based side-channel attacks, and earlier research on energy-, thermal-, reliability-, and aging-aware scheduling and reinforcement-learning-based optimization. My work spans algorithm design, mathematical modeling, large-scale simulation, and machine learning, and I implement and evaluate these systems in Python and C/C++ to bridge research ideas with production-quality systems software. I also bring industry experience from a software engineering internship, alongside teaching experience across a dozen undergraduate and graduate courses as a teaching assistant and lab instructor. I have authored 9 peer-reviewed publications spanning real-time scheduling, systems security, reliability, and reinforcement-learning-based optimization.

Education

George Mason University

Fairfax, VA

Ph.D. in Computer Science (GPA: 3.81/4.0)

August 2022 - Present (Expected 05/2028)

Sharif University of Technology

M.Sc. Computer Engineering - Computer Systems Architecture (GPA: 4.0/4.0)

2018 - 2021

Ferdowsi University of Mashhad

B.Sc. Computer Engineering - Hardware

2012 - 2017

Work Experience

George Mason University (Systems and Networking Group)

August 2022 - Present

Graduate Research Assistant

Fairfax, VA

- Developed large-scale C++ and Python simulation frameworks to evaluate real-time scheduling algorithms, grounded in concurrency, multi-threading, and synchronization.
- Designed and implemented security-aware scheduling techniques for real-time embedded systems to mitigate schedule-based side-channel attacks; published in **ISORC 2026** and **JSA 2023**.

George Mason University

August 2022 - Present

Graduate Teaching Assistant

Fairfax, VA

- Supported courses spanning Computer Systems Architecture, SWE for Real-Time Embedded Systems, Computer Communications and Networking, Database Systems, and Program Design (Java/Python/C), Held weekly recitations and office hours, providing feedback on code and assignments for 40+ undergraduate/graduate students.

Sharif University of Technology (Embedded Systems Research Laboratory)

2018 - 2021

Graduate Research Assistant

- Designed energy-, thermal-, reliability-, and aging-aware scheduling techniques for real-time embedded systems to improve efficiency, safety, and fault tolerance.
- Developed and analyzed system-level optimization techniques, and explored reinforcement learning-based approaches for real-time embedded systems.

Sharif University of Technology

2018 - 2021

Graduate Teaching Assistant

- Supported courses Fault-Tolerant Systems Design, Low Power Design, Digital Systems Design, Numerical Methods, Embedded Systems, Real-Time Systems, Logical Circuits Laboratory (as Instructor), and Digital Systems Design Laboratory (as Instructor), Held weekly office hours, providing feedback on code and assignments for 50+ undergraduate/graduate students.

Ferdowsi University of Mashhad

May 2016 - August 2016

Software Engineering Intern, Mini Computer Center (Peyman Nikkaran Tabaran Co.)

- Tested features for a zero client, mini computer, and thin client remote desktop application.
- Debugged and resolved connectivity/session handling issues affecting client deployments.

Ferdowsi University of Mashhad (Dependable Distributed Embedded Systems Laboratory)2014 - 2017

Undergraduate Research Assistant

- Analyzed the μ C/OS-III task scheduler in C for an ARM Cortex-M3-based real-time embedded system.

Ferdowsi University of Mashhad

2014 - 2017

Undergraduate Teaching Assistant

- Supported courses Logical Circuits, Machine and Assembly Languages, Computer Architecture, Held weekly office hours, providing feedback on code and assignments for 50+ undergraduate/graduate students.

Technical Skills

Programming Languages & Frameworks: C/C++, Java, Python, MATLAB, Verilog, VHDL, SQL, Bash, Embedded C, Linux Device Driver Development, JTAG, UART

Systems & Data: Git, MySQL, NumPy, Pandas

AI & Machine Learning: Reinforcement Learning, Genetic Algorithm

Simulation Tools: SLURM, CMake, Gem5, McPAT, HotSpot, Quartus, Xilinx, Design Compiler, Hspice, Pspice

Embedded/FPGA Platforms: Raspberry Pi, ARM Cortex, AVR, STM, NXP LPC, DE1, DE2, Spartan

Work Authorization: I am a permanent resident and do not need work authorization or visa support

Publications

S. Yari Karin, H. Aydin, and D. Zhu, "Task Splitting to Mitigate Schedule-Based Anterior Attacks in Real-Time Embedded Systems," *29th International Symposium on Real-Time Distributed Systems (ISORC)*, 2026.

S. Yari Karin, H. Aydin, D. Zhu, S. Drager, and M. Anderson, "Impact of Priority Assignment on Schedule-Based Attacks in Real-Time Embedded Systems," *Journal of Systems Architecture*, 2023.

S. Yari Karin, R. Siyadatzaheh, M. Ansari, and A. Ejlali, "Passive Primary/Backup-Based Scheduling for Simultaneous Power and Reliability Management on Heterogeneous Embedded Systems," *IEEE Transactions on Sustainable Computing*, 2022.

S. Safari, M. Ansari, H. Khdr, P. Gohari-Nazari, S. Yari Karin, A. Yeganeh-Khaksar, S. Hessabi, A. Ejlali, and J. Henkel, "A Survey of Fault-Tolerance Techniques for Embedded Systems from the Perspective of Power, Energy, and Thermal Issues," *IEEE Access*, 2022.

M. Ansari, S. Safari, S. Yari Karin, P. Gohari-Nazari, H. Khdr, M. Shafique, J. Henkel, and A. Ejlali, "Thermal-Aware Standby-Sparing Technique on Heterogeneous Real-Time Embedded Systems," *IEEE Transactions on Emerging Topics in Computing*, 2021.

M. Ansari, S. Yari Karin, S. Safari, and A. Ejlali, "Power Management to Meet Thermal Safe Power in Fault-Tolerant Embedded Systems," *TechRxiv*, 2021.

A. Yeganeh-Khaksar, M. Ansari, S. Safari, S. Yari Karin, and A. Ejlali, "Ring-DVFS: Reliability-Aware Reinforcement Learning-Based DVFS for Real-Time Embedded Systems," *IEEE Embedded Systems Letters*, 2021.

J. Saber-Latibari, M. Ansari, P. Gohari-Nazari, S. Yari Karin, A. M. H. Monazzah, and A. Ejlali, "READY: Reliability- and Deadline-Aware Power-Budgeting for Heterogeneous Multicore Systems," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2021.

S. Yari Karin, A. Sahraee, J. Saber-Latibari, M. Ansari, N. Rohbani, and A. Ejlali, "A Comparative Study of Joint Power and Reliability Management Techniques in Multicore Embedded Systems," *CSI/CPSSI International Symposium on Real-Time and Embedded Systems and Technologies (RTEST)*, 2020.

Awards & Honors

- **Student Travel Grant, ACM SIGMETRICS 2025** | *Accepted research poster on real-time systems security* (2025)
- **Certificate in Inclusive Teaching for STEM Graduate Students** | *Completed certified training on inclusive and effective STEM teaching practices* (2025)
- **Top 0.1%, Nationwide M.Sc. Entrance Exam** | *Ranked 30th out of $\approx$ 30,000 in Computer Engineering Exam* (2018)
- **Top 5 M.Sc. Student, Computer Systems Architecture** | *Ranked 5th out of 27* (2021)